Market Memory Structure and the Cost of Over-Differencing in ML Forecasting

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Abstract

We document two findings using 464 S&P 500 constituents over 2010–2025. First, the cost of differencing error is sharply asymmetric for modern machine learning: over-differencing irreversibly destroys predictive autocorrelation structure, while mild under-differencing is harmless because modern ML models are sufficiently flexible to learn through residual non-stationarity. Full differencing ($d = 1$) — the textbook prescription — produces out-of-sample forecast correlations indistinguishable from a coin flip across all memory regimes. Second, the walk-forward detectability boundary d^{LTM}, estimated via the finite-sample procedure of Bhatti (2026), reveals genuine and coherent structure in how equity return-generating processes store information — structure that varies meaningfully across tickers, across memory regimes, and through macroeconomic time. The theoretical foundation and estimation methodology behind d^{LTM} are developed in Bhatti (2026); this paper concerns only what that estimator reveals about equity market structure.

1 Introduction

The standard preprocessing recipe for financial ML is to difference price series to stationarity — for equities, that means log-returns ($d = 1$). This was sound advice when models were linear. Modern ML models are sufficiently flexible to handle mild non-stationarity without degradation. But they share one fundamental property with every other forecasting model: *information removed during preprocessing cannot be recovered.*

Fractional differencing with order $d \in (0, 1)$ attenuates long-run memory without fully eliminating it (Granger & Joyeux, 1980; Hosking, 1981). For $d = 1$ the operator reduces to the first difference and erases all autocorrelation structure beyond lag one. For $d < 1$, progressively more of the price-level memory is retained in the feature series.

This paper asks a single empirical question across 464 S&P 500 constituents over fifteen years: *what level of fractional differencing maximises out-of-sample ML forecast performance, and does the answer vary with the memory structure of the underlying series?* The answer to both parts is yes, and the mechanism is the same in each case.

Contribution 1: The cost of over-differencing is large, asymmetric, and universal. Full differencing ($d = 1$) destroys predictive signal across every memory regime we study. The median out-of-sample Pearson correlation under $d = 1$ is -0.0008 across 464 tickers, with exactly 50% positive — statistically indistinguishable from noise. By contrast, $d = 0.5$ achieves a median of $+0.0191$

with 76% of tickers positive, and ticker-specific d^{LTM} achieves +0.0126 with 70% positive. A 76% positive rate across 464 independent series over 15 years is statistically overwhelming; significance is established by the breadth of the cross-section, not individual-ticker magnitude. Modern ML models are robust to mild non-stationarity. They are not robust to destroyed information.

Contribution 2: d^{LTM} at 2,400-day granularity reveals genuine market memory structure. The walk-forward d^{LTM} is not a static preprocessing parameter — it is a slowly varying, cross-sectionally heterogeneous state variable that indexes how different equities store and dissipate information. At a 2,400-day estimation window, d^{LTM} partitions the cross-section into three regimes with markedly different distributional properties and different optimal differencing choices. This structure maps coherently onto macroeconomic regimes, as documented in Bhatti (2026), and different estimation granularities illuminate different temporal scales of market memory in ways that a single fixed window cannot capture.

The d^{LTM} estimator used throughout is not available from any standard econometric package and is not equivalent to any existing long-memory estimator — standard frequency-domain methods collapse to near-unit-root solutions on real equity data and provide no usable cross-sectional variation. Its theoretical properties, finite-sample calibration, and convergence analysis are established in Bhatti (2026).

2 Data and Experimental Design

Data. Daily log-prices for 464 S&P 500 constituents, January 2010 (2400 earlier dates used to train model for first prediction) through December 2025, adjusted for dividends and splits. Tickers with fewer than 2,400 trading days are excluded.

The d^{LTM} estimator. Following Bhatti (2026), d^{LTM} is estimated on a rolling 2,400-day window and re-estimated every 50 trading days in a strictly walk-forward manner — no future data enters any estimate. The procedure identifies the largest fractional differencing order at which hyperbolic ACF tail structure remains statistically distinguishable from noise, evaluated over lags 20–200 to isolate long-memory dependence from short-run GARCH effects, retaining only lags significant at the 5% level. Standard frequency-domain estimators applied to the same data produce estimates clustered in $[0.98, 1.02]$ for over 80% of tickers (Bhatti, 2026) — no usable cross-sectional variation.

Memory regimes. Tickers are classified by their median out-of-sample d^{LTM}: Low ($d^{\mathrm{LTM}} \leq 0.3$, mean-reverting, $n = 328$), Mid ($0.3 < d^{\mathrm{LTM}} \leq 0.7$, transitional, $n = 441$), and High ($d^{\mathrm{LTM}} > 0.7$, persistent, $n = 399$).

Forecasting experiment. Three independent preprocessing pipelines — d^{LTM}, $d = 0.5$, $d = 1$ — are compared using an identical model and feature structure, so that performance differences are attributable solely to the differencing choice. The forecasting horizon is one trading day ($h = 1$); the justification for this restriction is discussed below.

Feature construction. For each date t and differencing order d, the fractionally differenced series is

$$r_t^{(d)} = \Delta^d p_t = \sum_{j=0}^{J} w_j(d)\, p_{t-j}, \qquad w_0(d) = 1,$$

where $w_j(d)$ are the binomial coefficients truncated at $|w_j| < 10^{-4}$ and $p_t = \log P_t$. The feature vector at time t is the 10 most recent values of this series:

$$\mathbf{X}_t^{(d)} = \left[r_t^{(d)},\ r_{t-1}^{(d)},\ \ldots,\ r_{t-10}^{(d)}\right]^\top \in \mathbb{R}^{11}.$$

Prediction models. We evaluate two model variants for each pipeline. The **FD-direct** variant targets next-day returns directly from the fractionally differenced features:

$$\hat{y}_{t+1} = f_{\text{ret}}\big(\mathbf{X}_t^{(d)}\big).$$

The **FD-inversion** variant first predicts the one-step-ahead value of the fractionally differenced series,

$$\hat{r}_{t+1}^{(d)} = f_{\text{fd}}\big(\mathbf{X}_t^{(d)}\big),$$

and then maps this prediction back to return space via exact inversion (described below). A **raw-return benchmark** uses lagged log-returns directly as features and targets, with no fractional differencing. All models use a **Histogram-Based Gradient Boosted Regressor** (`HistGradientBoostingRegressor`, scikit-learn, max depth 3), retrained every 50 days on an expanding window with a minimum burn-in of 1,600 days. The architecture is held fixed across all pipelines to isolate the effect of the differencing choice.

Exact inversion to return space. Since $w_0(d) = 1$, the fractional differencing operator can be inverted exactly at $h = 1$. Substituting the model's FD-level prediction into the inversion formula:

$$\hat{p}_{t+1} = \hat{r}_{t+1}^{(d)} - \sum_{j=1}^{J} w_j(d)\, p_{t+1-j},$$

where all lagged log-prices p_{t+1-j}, $j \geq 1$, are observed at decision time t. The predicted return is then

$$\hat{r}_{t+1}^{\text{inv}} = \hat{p}_{t+1} - p_t.$$

This inversion introduces no approximation error at $h = 1$; numerical verification confirms reconstruction errors below 10^{-10} across all assets and time periods. Without this step, FD-level predictions and return-space predictions occupy incommensurable spaces and cannot be compared on identical grounds.

Why $h = 1$ only. The inversion above is exact precisely because all lagged prices entering the formula are observed at decision time t. For $h > 1$, predicting $r_{t+h}^{(d)}$ requires intermediate log-prices $p_{t+1}, \ldots, p_{t+h-1}$ that are unobserved at time t, introducing approximation error that compounds across steps. We restrict to $h = 1$ to keep the inversion exact and the comparison across pipelines clean. Multi-step extension via iterated one-step prediction or direct horizon models is left for future work.

The primary evaluation metric is the per-ticker Pearson correlation between $\hat{r}_{t+1}^{\text{inv}}$ and the realised log-return $y_{t+1} = p_{t+1} - p_t$, computed over the full out-of-sample period. All three pipelines are evaluated in the same return space.

3 Results

3.1 Full Cross-Section

Figure 1 shows the kernel density of per-ticker forecast correlations across all 464 tickers.

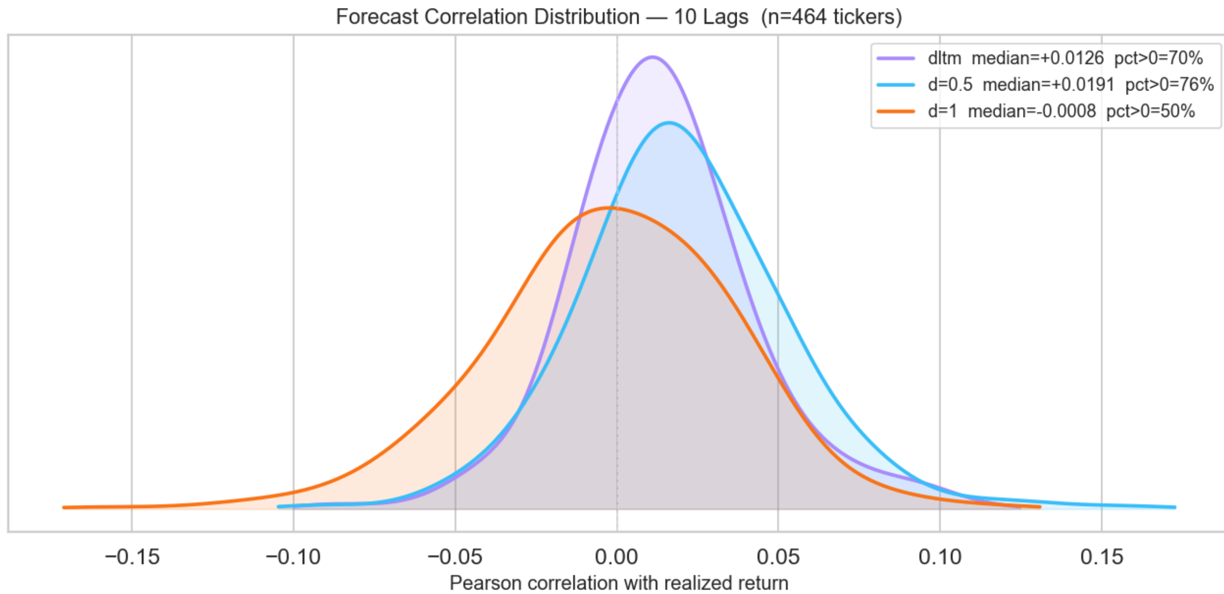

Figure 1: Kernel density of per-ticker Pearson forecast correlations, 464 S&P 500 tickers, 10-lag features, $h = 1$, 2010–2025. Purple: d^{LTM} (median +0.0126, 70% positive). Cyan: $d = 0.5$ (median +0.0191, 76% positive). Orange: $d = 1$ (median −0.0008, 50% positive).

The $d = 1$ distribution is centred at zero and symmetric — the signature of preprocessing that has eliminated the signal entirely. The d^{LTM} distribution is the most concentrated of the three, with the sharpest peak, but its pooled median trails $d = 0.5$. The regime analysis explains why.

3.2 Regime-Stratified Results

Figure 2 and Table 1 tell the full story.

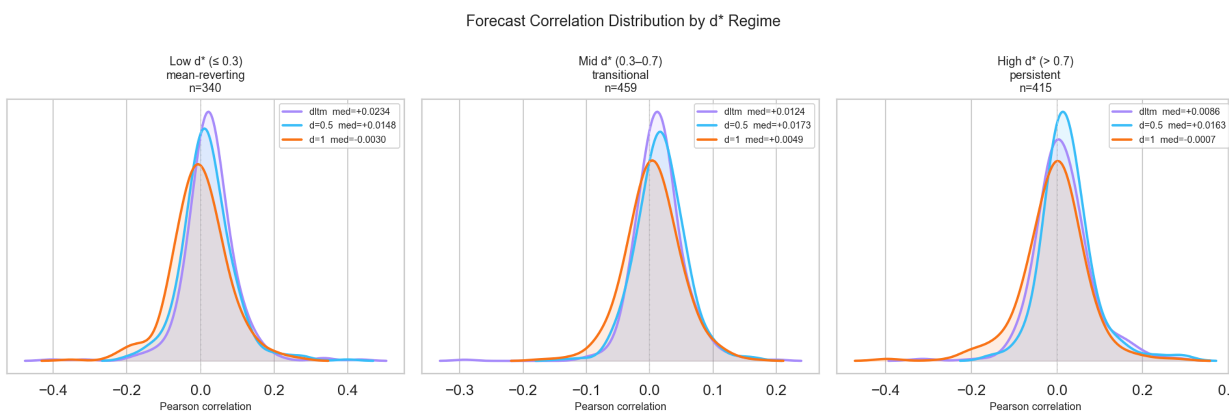

Figure 2: Forecast correlation distributions by d^{LTM} memory regime. *Left:* Low d^{LTM} (≤ 0.3, mean-reverting, $n = 328$). *Centre:* Mid d^{LTM} (0.3–0.7, transitional, $n = 441$). *Right:* High d^{LTM} (> 0.7, persistent, $n = 399$). Purple: d^{LTM}. Cyan: $d = 0.5$. Orange: $d = 1$.

Table 1: Out-of-Sample Forecast Correlation by Regime and Differencing Method

Regime	Metric	d^{LTM}	$d = 0.5$	$d = 1$
Full Sample ($n = 464$)	Median corr.	+0.0126	+0.0191	−0.0008
	Pct > 0	70%	76%	50%
Low d^{LTM} (≤ 0.3, $n = 328$)	Median corr.	**+0.0235**	+0.0131	−0.0066
	Interpretation	d^{LTM} wins: $d = 0.5$ already over-differences mean-reverting series		
Mid d^{LTM} (0.3–0.7, $n = 441$)	Median corr.	+0.0121	**+0.0173**	+0.0042
	Interpretation	$d = 0.5$ and d^{LTM} similiar performance		
High d^{LTM} (> 0.7, $n = 399$)	Median corr.	+0.0086	**+0.0163**	−0.0005
	Interpretation	$d = 0.5$ wins: d^{LTM} tracks stationarity but not useful for this use case		

Note: Bold denotes highest median correlation within each regime. All correlations are out-of-sample Pearson with realised log-returns.

The regime results form a single coherent argument. In the Low d^{LTM} regime: these are genuinely mean-reverting series and $d = 0.5$ already over-differences them. d^{LTM} wins by correctly estimating d well below 0.5. In the High d^{LTM} regime, the conservative $d = 0.5$ wins by differencing less than the estimated optimum.

In every regime, the winner is whichever method differences *least* relative to the true process. The cost function is asymmetric and the asymmetry runs in one direction only: over-differencing carries a large irreversible penalty; under-differencing carries less for modern ML models.

4 What d^{LTM} Reveals About Market Memory

The regime structure above is only interpretable if d^{LTM} is measuring something real about the underlying return-generating process. It is.

At 2,400-day granularity, the three memory regimes display starkly different distributional properties. Raw return excess kurtosis rises monotonically from 3.90 (Low) to 9.62 (High); after volatility adjustment it collapses to 0.72 and 0.96 respectively, confirming that the non-normality is driven by volatility clustering rather than structural non-stationarity. Negative skewness deepens from -0.14 to -0.46 as d^{LTM} rises, consistent with asymmetric shock absorption in more persistent processes. These are systematic differences in how equities store and dissipate information, and d^{LTM} is indexing them reliably.

Table 2: Distributional Properties of Log-Returns by d^{LTM} Regime

Panel A: Raw Returns

Regime	n	Med. Skew	Med. Ex. Kurt.	% Leptokurtic	JB Reject %
Low d^{LTM} (≤ 0.3)	328	-0.144	3.90	96.3%	89.6%
Mid d^{LTM} (0.3–0.7)	441	-0.201	8.00	100.0%	99.5%
High d^{LTM} (> 0.7)	399	-0.461	9.62	99.7%	96.7%
Full Sample	446	-0.379	10.98	100.0%	100.0%

Panel B: Volatility-Adjusted Returns

Regime	n	Med. Skew	Med. Ex. Kurt.	% Leptokurtic	JB Reject %
Low d^{LTM} (≤ 0.3)	328	-0.096	0.72	93.3%	68.9%
Mid d^{LTM} (0.3–0.7)	441	-0.118	0.90	99.8%	97.1%
High d^{LTM} (> 0.7)	399	-0.185	0.96	97.0%	91.5%
Full Sample	446	-0.151	0.93	100.0%	99.6%

Notes: JB Reject % is the fraction of tickers for which the Jarque–Bera test rejects normality at the 5% level. Excess kurtosis is reported (normal = 0).

The cross-sectional d^{LTM} distribution is also time-varying in macroeconomically coherent ways: it troughs after the GFC as prices became dominated by mean-reversion, recovers through the QE bull market as persistence rebuilds, and declines again after COVID and the Fed hiking cycle. The full time-series analysis — including convergence across estimation windows from 800 to 3,200 days, temporal stability, and macroeconomic event attribution — is in Bhatti (2026).

A key implication for future work: different estimation granularities illuminate different temporal scales of this memory landscape. Short windows capture fast local regime shifts; long windows reveal structural multi-year patterns. Each granularity provides a distinct view of market memory structure that a single fixed window cannot capture. The framework for a systematic multi-granularity analysis — and the estimator to conduct it — exists in Bhatti (2026).

For the distributional analysis in Section 4, the sample reduces to $n = 446$: assets with fewer than 100 out-of-sample predicted days are additionally excluded, as the distributional statistics are computed on the same filtered dataset used in the forecasting experiment.

5 Conclusion

Two things are true simultaneously, and together they matter.

Over-differencing is the dominant source of preprocessing-induced forecast degradation in ML pipelines on equity time series. The standard recipe — difference to stationarity, apply model — is actively harmful when the model is a modern ML regressor.

While the long-memory index d_{LTM} may admit richer or more targeted exploitation strategies, our objective here is not to optimize information extraction but to conduct a controlled comparison. Accordingly, we hold the forecasting architecture fixed and use a standard machine learning model to isolate the impact of differencing alone.

The d^{LTM} estimator is not just a better preprocessing tool. At 2,400-day granularity it reveals genuine, time-varying, cross-sectionally heterogeneous structure in how equity return-generating processes store information — structure with measurable forecasting consequences and coherent macroeconomic interpretation.

Both findings depend on having a d^{LTM} estimator that works on real financial data — and on the same estimator producing usable cross-sectional variation where standard methods do not. The construction, theoretical properties, finite-sample calibration, and convergence analysis are in Bhatti (2026).

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